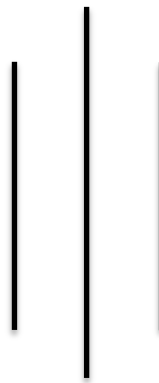


Report on Wastewater Treatment Plan of Rupa Rural Municipality, Kaski, Nepal

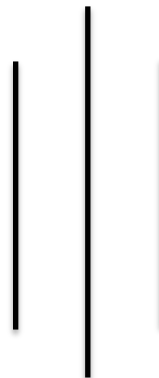


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Lastly, my sincere appreciation goes to my peers and friends who supported me during this project. Their collaboration and encouragement were truly inspiring.

PREFACE

This report on wastewater treatment for Rupa Rural Municipality, Ward 5 and 6, aims to explore the current sanitation practices and offer feasible solutions for improved sewage management. With the increasing population and limited infrastructure in rural areas, the need for an effective wastewater treatment system has become a pressing issue. Proper management of wastewater not only ensures environmental sustainability but also promotes public health and improves the quality of life for the residents.

The objective of this mini project is to explore the existing sanitation infrastructure, identify key issues, and propose a feasible and efficient solution for wastewater treatment that aligns with modern engineering practices. The report delves into both technical and practical aspects of sanitary engineering, emphasizing the importance of sustainable and context-specific solutions for rural municipalities.

This study has been conducted as part of sanitary engineering under the Bachelor of Civil Engineering program at the Institute of Engineering (IOE), Western Regional Campus, Pokhara. Through this project, we aim to offer meaningful insights and contribute to ongoing efforts for improving sanitation in the region, ensuring that the solutions we propose are not only technically sound but also environmentally and socially responsible.

This report is a result of collective efforts, research, and practical study, and it seeks to lay the foundation for a cleaner and more sustainable future for Rupa Rural Municipality.

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1. INTRODUCTION

Wastewater is a byproduct of human and animal activities, often containing harmful substances that pose risks to the environment and public health. Domestic and industrial waste discharges are major contributors to environmental degradation, especially in regions with increasing population density. Insufficient sanitation and improper wastewater management lead to the spread of communicable diseases, primarily those caused by human and animal waste. Both solid and liquid wastes should be properly collected and conveyed for treatment and disposal.

In many rural areas of Nepal, sanitation infrastructure remains underdeveloped. Over 80% of wastewater generated in less developed regions is discharged untreated into nearby water bodies, leading to significant environmental contamination. In Rupa Rural Municipality, Wards 5 and 6, which is our study area, the lack of a proper sewage network and treatment facility has resulted in the direct discharge of untreated wastewater into natural water bodies, creating pollution and increasing the risk of waterborne diseases. Open defecation and improper disposal of sewage are prevalent issues, leading to soil and water contamination, which further threaten public health and the surrounding ecosystem.

2. BACKGROUND

2.1 Location and Topography

Rupa Rural Municipality (28.1447°N, 84.0470°E) is situated in the Kaski District of Gandaki Province, Nepal. It lies within the mid-hill region of the country, characterized by undulating terrain, terraced farmlands, and scattered settlements. The area is intersected by small streams and rivulets, which contribute to its natural water resources and agricultural productivity. On 12 March 2017, the government of Nepal implemented a new local administrative structure in which VDCs have been replaced with municipal and Village Councils. Rupa is one of these 753 local units. It has been divided into 7 wards and has an area of 94.8 km². The project area, consisting of two wards, 5 and 6 has the total area of 35.34 km² and the population of 5590.

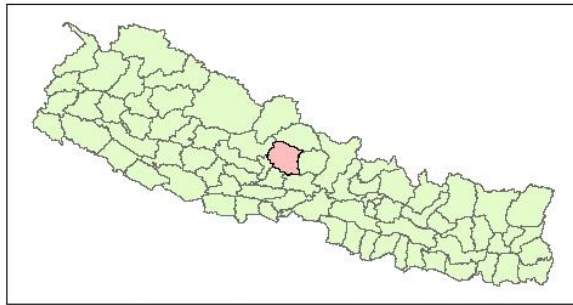


Fig- Location of Kaski on Nepal

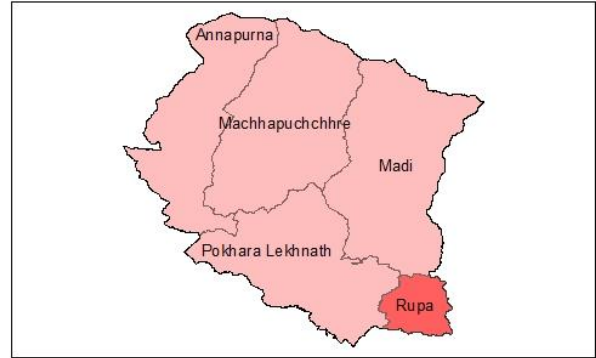


Fig- Kaski District

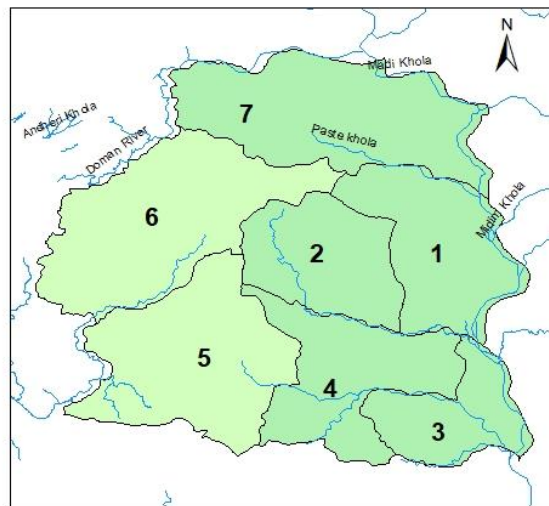


Fig- Rupa Rural Municipality



Figure 1-Location of Rupa Rural Municipality

2.3 Climatic Conditions

Rupa experiences a subtropical highland climate, which is characterized by moderate temperatures and distinct seasonal variations. The average annual temperature ranges between 15°C to 25°C (59°F to 77°F). Humidity levels are high during the monsoon season due to heavy rainfall, in other seasons, humidity is relatively moderate. The region's climatic conditions, characterized by a heavy monsoon season and a mild winter, create challenges in ensuring consistent water supply throughout the year.

The average rainfall intensity in the region is approximately 3 mm per hour, with peak intensities during the monsoon season reaching up to 30–40 mm per hour during extreme weather events. The annual precipitation in the area is around 3,500 mm, with over 80% of it occurring during the monsoon season.

2.3 Economic and Development Conditions

Rupa Rural Municipality's economy relies heavily on agriculture, with crops like rice, maize, and millet as key sources of livelihood. Livestock farming, including cattle and poultry, supports the local economy, alongside small businesses such as retail shops. However, agricultural productivity faces challenges due to weather-related risks and limited market access.

Development efforts focus on improving infrastructure, such as road connectivity and access to services. Upgrades to educational and healthcare facilities are ongoing, while governance and institutional initiatives are working to enhance public well-being. Despite progress, further investment in sustainable infrastructure and economic diversification is necessary for long-term growth.

3. PROBLEM IDENTIFICATION

Rupa Rural Municipality, particularly in wards 5 and 6, faces significant challenges in managing wastewater and ensuring proper sanitation. The absence of a structured sewage system results in untreated wastewater being discharged into open fields, streams, and rivers, causing soil and water contamination. This practice not only pollutes the environment but also contributes to the spread of waterborne diseases, posing a threat to public health. Additionally, the growing population and lack of sanitation awareness further promoted the issue. Inadequate infrastructure, mix with the lack of waste treatment facilities really needs urgent intervention to address these sanitation and health concerns.

4. OBJECTIVE OF THE PROJECT

The overall objective of this project is to establish a sustainable wastewater management system in Rupa Rural Municipality, wards 5 and 6, by connecting locations with gravity-fed pipelines. The project aims to improve sanitation, prevent pollution, and enhance public health through an effective sewage treatment system, while raising awareness about proper sanitation practices and supporting future infrastructure development.

5. QUANTITY ESTIMATION OF SEWAGE

Sanitary sewage consists of wastewater generated from various domestic, industrial, and public activities. Water supplied for domestic use, after serving its purpose, is discharged into sewers. Similarly, industries and public places such as schools, hotels, and hospitals generate wastewater, a portion of which enters the sewer system after use. Additionally, infiltration of groundwater through leaky joints and rainwater entering through cracks in sewers during the rainy season further adds to the wastewater load.

Accurate estimation of sewage discharge is crucial for the hydraulic design of sewers. Underestimation can result in inadequate sewer capacity, while overestimation leads to oversized, uneconomical sewers and potential deposition due to insufficient self-cleansing velocity. Proper estimation ensures efficient and sustainable sewerage design.

5.1 Design Period

The design period refers to the future duration for which the capacities of various components of a sewerage system are planned. It is influenced by factors such as ease of expansion, availability of investment, anticipated population growth, hydraulic constraints, and the lifespan of materials and equipment.

Sewers are typically designed for peak flows expected over 25 to 50 years. Considering the durability of materials like concrete pipes, which are more prone to damage compared to steel, and the challenges in reconstruction and expansion, a design period of 30 years has been selected for this project to ensure long-term functionality and efficiency.

Hence, Survey year: 2025 AD

Base Period: 2 yrs.

Design year: **2057 AD** Design Period: **30 yrs.**

5.2 Water Demand

5.2.1 Population Forecasting

(Source: CBS)

Census Year (AD)	Population	Increase in population	Percentage increase (%)	Decrease in percentage increase (%)
2001	4981			
2011	5331	350	7.02 %	
2021	5590	259	4.8 %	2.22 %

Average decrease in percentage increase in population per decade $= r' = 2.22 \%$

$n = 3.6$ (from 2021 to 2057)

Percentage increase in population in last decade $= r_t = 4.8 \%$

By using changing rate of increase method,

The prospective population at the end of the year 2041 is given by

$$\begin{aligned}
 P_{2057} &= P_{2021} * (1 + (r_t - r') / 100) * (1 + (r_t - 2 * r') / 100) * (1 + (r_t - 3 * r') / 100) * \\
 &\quad 1 + (r_t - 0.6 * r') / 100 \\
 &= 5590 * (1 + (4.8 - 2.22) / 100) * (1 + (4.8 - 2 * 2.22) / 100) * (1 + (4.8 - 3 * 2.22) / 100) * \\
 &\quad 1 + (4.8 - 0.6 * 2.22) / 100 \\
 &= 5844
 \end{aligned}$$

Hence, the population by the end of 2057 is 5844.

1) Domestic Demand (DD)

Consumption in litres per capita per day = 65 lpcd

$$DD = P_{2057} * 65$$

$$DD = 5844 * 65$$

$$DD = 379860 \text{ lpd} = 379.86 \text{ m}^3/\text{day}$$

2) Livestock Demand (LD)

(Source: CBS)

Size of animals	Number of animals	Consumption per day	Demand (m ³ /day)
Large	2357	45 l / animal / day	106.065
Medium	5371	20 l / animal / day	107.42
Small	15274	20 l / 100 animals / day	3.054
Total	23002		216.539 m ³ /day

$$20\% \text{ of } DD = 0.2 * 374.075 = 74.815 \text{ m}^3/\text{day}$$

Taking the minimum of two values, LD = 74.815 m³/day

3) Commercial Demand (CD)

(Source: CBS, rural municipality office)

Item	Number	Consumption	Demand (m ³ /day)
Day Scholars	853	10 lpcd	8.53
Offices	3	700 l / day/ office	2.1
Hospital with beds	1(15 beds)	500l/bed/day	7.5
Health clinics	2	2500l/day	5
Hotel with beds	3(142 beds)	200l / bed/day	28.4
Hotel without beds/ Restaurants	11	500l/day	5.5
Total			57.03

Hence, CD = 57.03 m³/day.**4) Public / Municipal Demand (PD)**

Public demand is taken into consideration for the fact that there are few public parks and some public view spots around the area.

PD = 5 % of TD (Let the total demand (TD) be 'x')

$$= 0.05 x$$

5) Fire Demand (FD)

The overall fire risk and incidence rates in Rupa are low. The infrastructure for firefighting is not developed. Hence, being practical / cost-effective, 5 % of total demand is taken for fire demand.

FD = 5 % of TD

$$= 0.05 x$$

6) Loss and Wastage Demand (LW)

LW = 15 % of TD

$$= 0.15 x$$

Total Demand (TD)

$$DD + LD + CD + FD + PD + LW = TD$$

$$379.86 + 74.815 + 57.03 + 0.15x + 0.05x + 0.05x = x$$

Therefore, $x = 682.273 \text{ m}^3/\text{day} = \text{Total demand}$

$$= 0.682 \text{ million litres per day}$$

$$= 28428.042 \text{ litres/hr}$$

$$= 28.4 \text{ m}^3/\text{hr}$$

Hence, the total demand for design year is **0.00789 m³/s**.

5.3 Dry weather flow

Generally, 75 to 80% of accounted water supplied is considered as quantity of sewage produced.

$$\text{Average quantity of DWF} = 0.8 * 0.00789 \text{ m}^3/\text{s} \quad [80\% \text{ taken.}]$$

$$= 0.00631 \text{ m}^3/\text{s}.$$

$$\text{Peak quantity of DWF} = P.F * 0.00631 \text{ m}^3/\text{s}$$

$$= 2.5 * 0.00631 \text{ m}^3/\text{s} \quad [\text{Let us take peak factor 2.5.}]$$

$$= 0.01578 \text{ m}^3/\text{s}.$$

5.4 Storm Water Flow

The commonly used method for estimating stormwater runoff is the **Rational Method**, which is simple and effective for small watersheds or urban areas.

Rational Method Formula

The Rational Method is expressed as:

$$Q = CIA/360$$

Where:

- Q: Peak runoff (m³/s)
- C: Runoff coefficient (dimensionless), which depends on land use, surface characteristics, and soil type
- I: Rainfall intensity (mm/hr) for the design return period and storm duration
- A: Catchment area (hectares)

The factor 360 is a unit conversion constant.

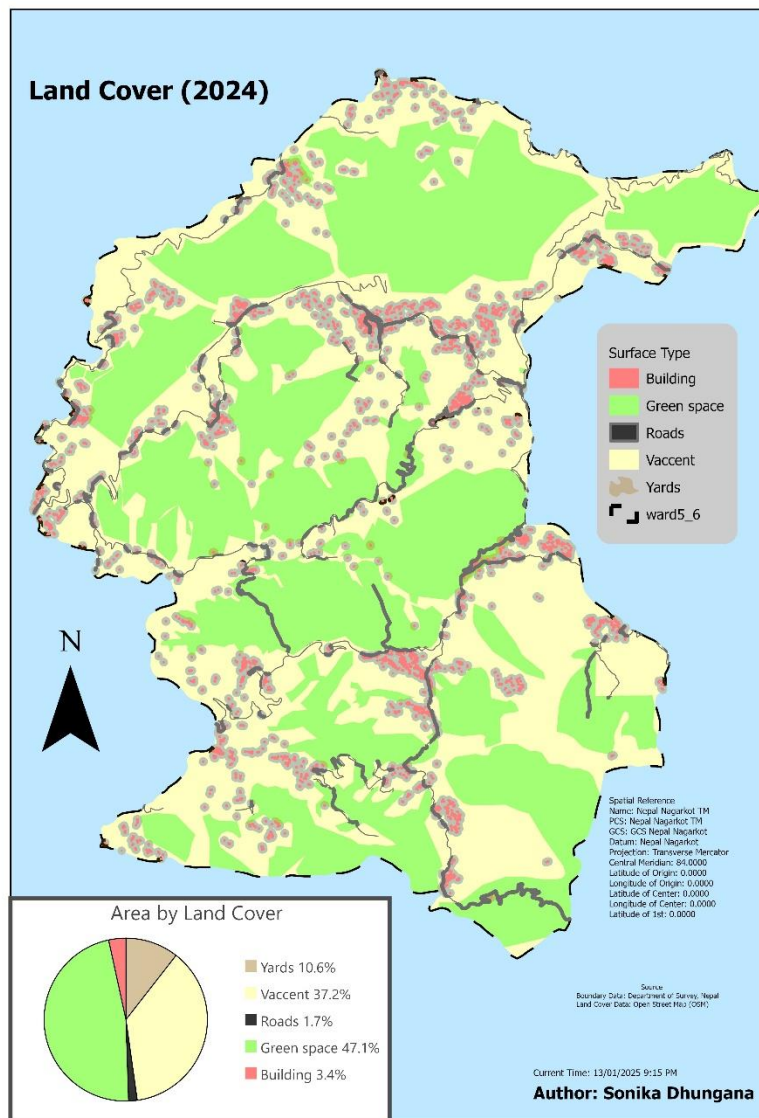


Figure 2-Land Cover map of Projected Area

The provided data corresponds to the year 2024. To estimate the conditions for the design year, the percentage of buildings is projected to increase, replacing a proportion of the vacant plots. This adjustment allows for a more accurate estimation based on anticipated future development.

We have, Area (A) = $35.34 \text{ km}^2 = 3534 \text{ ha}$.

Based on general climatic data for Kaski District, the region experiences an average annual rainfall of approximately 3,823 mm, with rainfall occurring on around 186 days each year. Assuming an average of 6 hours of rainfall per day, the estimated rainfall intensity for Rupa Rural Municipality, Ward 5 and 6, is approximately 3.44 mm per hour (Source: Kaski District Climate Study, 2025).

Rainfall Intensity (i) = 3.44 mm/hr

For C:

Types of Surfaces	% Area	Runoff Coefficient
Yards	10.6	0.5
Vacant Plots	27.2	0.3
Roads	1.7	0.8
Green Spaces (Gardens)	47.1	0.2
Buildings (Roofs)	13.4	0.85

$$\text{Average Runoff Coefficient (C)} = [10.6*0.5+27.2*0.3+1.7*0.8+47.1*0.2+13.4*0.85]/100 \\ = 0.383$$

So,

$$\text{Wet Weather Flow (WWF)} = C_i A / 360 = [0.383 * 3.44 * 3534] / 360 = 12.9336 \text{ m}^3/\text{s}$$

Calculation for only yards, roofs and pavements,

$$C = [10.6*0.5+1.7*0.8+13.4*0.85]/25.7 \\ = 0.702$$

$$\text{Partial WWF} = C_i A / 360 = [0.702 * 3.44 * (0.257*3534)] / 360 = 6.0924 \text{ m}^3/\text{s}$$

5.5 Sewage Quantity Calculation

The **partial drainage system** is chosen for Rupa Rural Municipality to manage surface runoff efficiently, reduce flooding risk, and provide a cost-effective solution for areas with varied topography, scattered households and limited infrastructure.

For partial Separate System:

$$\text{Quantity of Sewage in Sewer} = Q_{\text{partial sewer}} = (0.01578 + 6.0924) \text{ m}^3/\text{s}. \\ = 6.10818 \text{ m}^3/\text{s}.$$

$$\text{Quantity of Storm sewage in drain} = Q_{\text{partial drain}} = (12.9336 - 6.0924) \text{ m}^3/\text{s} \\ = 6.8412 \text{ m}^3/\text{s}$$

6. DESIGN OF SEWER

6.1 Hydraulic Design

Minimum and Maximum Velocity: The minimum velocity of 0.6 and 0.75 m/s at the peak flow is recommended for separate flow and combined/storm water sewer respectively.

The maximum velocity of sewage flow in sanitary sewer system should not be greater than 3m/s.

Shape and Type of Sewer:

Closed Circular for Sanitary Sewer

Open Rectangular or Trapezoidal for Storm Water Drainage.

6.2 Sewer Sizing and Alignment

For main Sewer Line (Line A):

Considering the circular sewer carrying (2/3)rd depth at peak discharge/ maximum discharge.

longitudinal slope of 1:400

and manning's coefficient(N)=0.012 for concrete pipes.

Here, when sewer run partially full,

$$d/D=2/3=0.667$$

$$\cos(\theta/2) = (1-2d/D) = -1/3,$$

$$\theta = 218.94$$

Now, calculated flow area and wetted perimeter.

$$a/A = [(\theta/360) - \sin(\theta)/2\pi]$$

$$= ((218.94/360) - \sin(218.94)/360)$$

$$a/A = 0.61$$

$$a = 0.61 \cdot [\pi \cdot D^2 / 4] \quad \text{So, } a = 0.55621 D^2$$

$$\text{And } r/R = (1 - 360 \cdot \sin(\theta)) / (2\pi\theta)$$

$$= 1 - 360 \cdot \sin 218.94 / (2\pi 218.94)$$

$$= 1.165$$

$$r = 1.65 \cdot D / 4 = 0.2911 D$$

Peak discharge(Q)= 0.0163892 m³/s.

Using Manning's Equation,

$$Q = (1/N) * a * r^{2/3} * s^{1/2}$$

$$6.10818 = (1/0.012) * 0.55621 D^2 * (0.2911D)^{2/3} * (1/400)^{1/2}$$

∴ D = 1.958 m = 2000 mm (According to RCC concrete standards available in Nepal.)

Check for Cleansing Velocity,

$$\text{Velocity (v)} = (1/N) * r^{2/3} * s^{1/2} = 2.81 \text{ m/s } (>0.75\text{m/s and } < 3\text{m/s}). \text{ OK.}$$

At minimum discharge,

Let minimum discharge = q = Average flow / 3

$$= (\text{Peak flow} / 2.5) / 3$$

$$= Q / 7.5$$

Then,

$$q/Q = 1/7.5 = [(\theta/360) - \sin(\theta)/2\pi] [(1-360*\sin(\theta)/(2\pi\theta))]^{2/3}$$

Solving the above eq, $\theta = 118.148$.

Also,

$$v/V = [(1-360*\sin(\theta)/(2\pi\theta))]^{2/3}$$

$$v = [(1-360*\sin(118.148)/(2\pi * 118.148))]^{2/3} * 2.81$$

$$= 1.66 \text{ m/s } > 0.4 \text{ m/s OK.}$$

For Sub-mains (Sewer Lines B and C):

Two sub-main sewers were added along with the main sewer in the sewage plan due to the scattered distribution of households. The area lacks a dense, grid-like settlement pattern, making it inefficient to rely solely on a single main sewer. Sub-main sewers help in efficiently collecting wastewater from dispersed households and directing it to the main sewer, ensuring an organized flow and reducing excessive pipe length and excavation costs.

For Submain Sewer Line B:

$$B = B1+B2+B3$$

The sub-main sewer line **B** has been divided into three segments: **B1, B2, and B3**, each designed to accommodate varying sewage inflows based on distinct population considerations. The flow distribution is structured as follows:

	B1	B2	B3
Length	2575.21m	2487.481m	2060.278m
Approx. Houses	394	632	789
Approx. Population	1828	2925	3565
DWF_{max}	0.00464 m ³ /s	0.0074m ³ /s	0.00928 m ³ /s
Q_{partial sewage}	1.595 m ³ /s	2.55m ³ /s	3.19 m ³ /s
Q_{partial drain}	2.011 m ³ /s	3.218 m ³ /s	4.022 m ³ /s

The houses and population approximate were analysed by statistics and GIS.

The wastewater quantity values were analysed by the same process as in the main sewer.

Material for pipes used: Concrete

Then,

Using manning's formula for max discharge at 2/3 D and checking the cleansing velocity, we get

$$D_{\text{line B1}} = 1.121\text{m} = 1200\text{mm}$$

$$\text{Slope} = 0.003$$

$$D_{\text{line B2}} = 1.336\text{m} = 1400\text{mm}$$

$$\text{Slope} = 0.003$$

$$D_{\text{line B3}} = 1.4541\text{m} = 1500\text{ mm}$$

$$\text{Slope} = 0.003$$

For Submain Sewer Line C:

$$C = C1 + C2$$

The sub-main sewer line **C** has been divided into two segments: **C1 and C2**, each designed to accommodate varying sewage inflows based on distinct population considerations. The flow distribution is structured as follows:

	C1	C2
Length	1937.467 m	1334.692 m
Approx. Houses	331.8	553
Approx. Population	1214	2024
DWF_{max}	0.0039 m ³ /s	0.0065m ³ /s
Q_{partial sewage}	1.5108 m ³ /s	2.518m ³ /s
Q_{partial drain}	1.691 m ³ /s	2.819 m ³ /s

The houses and population approximate were analysed by statistics and GIS.

The wastewater quantity values were analysed by the same process as in the main sewer.

Material for pipes used: Concrete

Then,

Using manning's formula for max discharge at 2/3 D and checking the cleansing velocity, we get

$$D_{\text{line C1}} = 1.098\text{m} = 1200\text{mm}$$

$$\text{Slope} = 0.003$$

$$D_{\text{line C2}} = 1.3306\text{ m} = 1400\text{ mm}$$

$$\text{Slope} = 0.003$$

6.3 Conclusion of Design

	Main Sewer Line A	Sub-Main Sewer Line B B1 B2 B3			Sub-Main Sewer Line C C1 C2	
Material Used	Concrete	Concrete			Concrete	
Slope	0.004	0.003			0.003	
Diameter	2000mm	1200mm	1400mm	1500mm	1200mm	1400mm
Length	1272.63 m	2575.21 m	2487.48 1m	2060.278 m	1937.467 m	1334.692 m

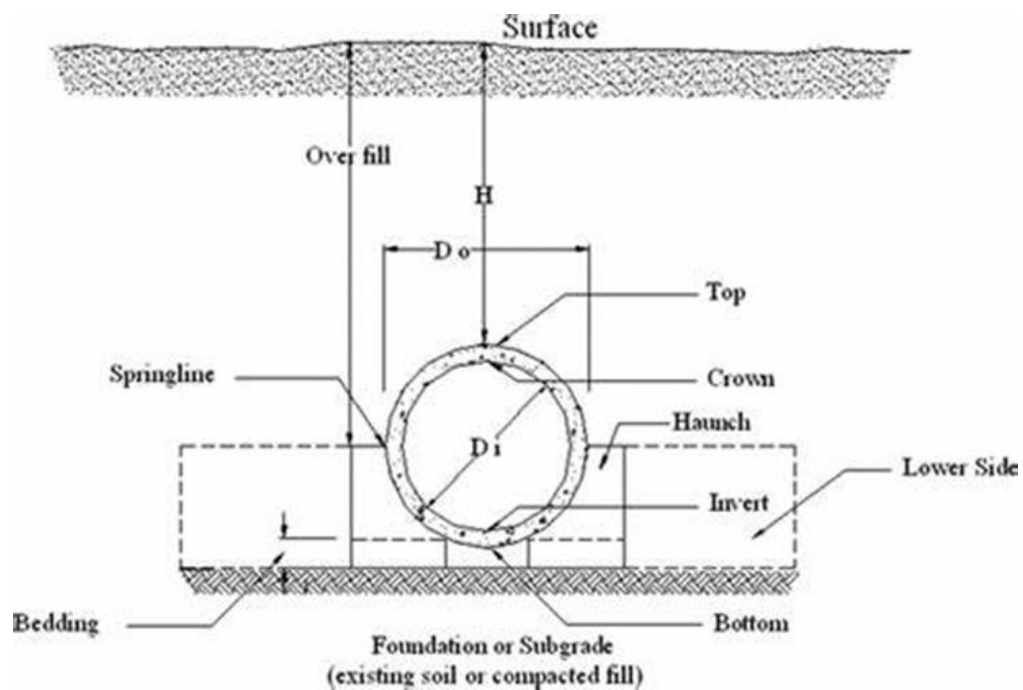


Figure 3-Circular Sewer

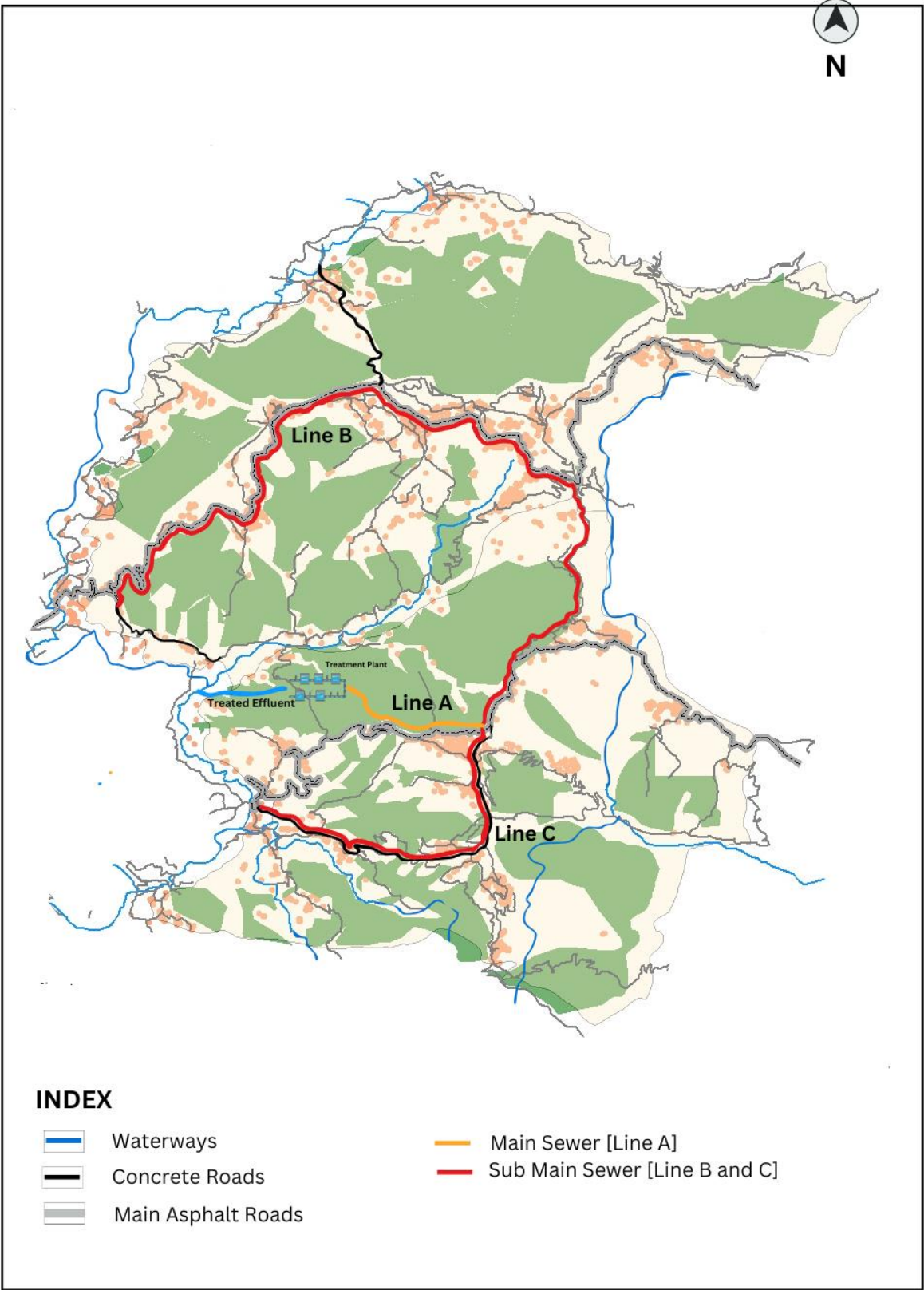


Figure 4-Main Sewer and Sub Main Sewer Layout Plan

7. STORM WATER DRAINS

A portion of the collected stormwater will be utilized for irrigation purposes, supporting agricultural activities in the region. Some of the runoff will be directed toward forested areas to enhance groundwater recharge and maintain ecological balance. The remaining stormwater will be conveyed through storm drains and discharged into the nearest natural waterway, ensuring proper drainage and minimizing waterlogging. The system is designed with appropriate slopes, alignments, and regular maintenance strategies to enhance its efficiency and sustainability.

The stormwater drains will be constructed using reinforced concrete or masonry, with trapezoidal or rectangular cross-sections for efficient flow. In areas where infiltration is beneficial, unlined channels will be used to enhance groundwater recharge and reduce surface runoff. Covered drains with inlets will be used in high-runoff areas. Regular maintenance, including desilting and debris removal, will ensure optimal performance and prevent blockages.

8. SEWER APPURTENANCES

Sewer appurtenances are essential components of a sewerage system, ensuring its efficient operation, maintenance, and longevity. The key sewer appurtenances considered for this rural municipality are discussed below:

7.1 Manholes

Manholes are provided at regular intervals to facilitate inspection, cleaning, and maintenance of the sewer system. Manholes are located at each sewer junction or change of sewer direction, diameter or gradient.

- Spacing between manholes is maintained between 30-50 meters.
- Constructed using brick masonry or precast concrete rings for durability.
- Equipped with proper covers to prevent Odor and contamination.

7.2 Drop Manholes

Drop manholes are used in areas where there is a significant change in elevation between incoming and outgoing sewers. They help in preventing excessive velocity, which can cause scouring of sewer pipes.

7.3 Flushing Tanks

Flushing tanks are used in areas with low water flow to maintain self-cleansing velocity in sewers.

- Automatic or manually operated tanks can be installed in critical locations.
- Helps in preventing blockages and buildup of solid waste.

7.4 Grease Traps

Grease traps are installed to capture fats, oils, and grease (FOG) from wastewater before they enter the sewer system.

- Essential for households and businesses generating high grease content.
- Prevents sewer blockages and environmental pollution.

7.5 Ventilating Shaft

Sewer vents ensure proper ventilation of gases within the sewer system, preventing the buildup of harmful gases like methane, carbon dioxide and hydrogen sulphide. It is strategically located at key junctions and prevents foul Odors and improve overall sewer efficiency.

9. TREATMENT OF SEWAGE

The wastewater treatment is performed to reduce the objectionable properties of water carrying waste and make it less dangerous for human and environment. Degree of treatment depends upon the treatment methods used and requirement of final effluent. Wastewater treatment process also make effluent safe for disposal and reduces its negative impacts on surrounding environment and human health.

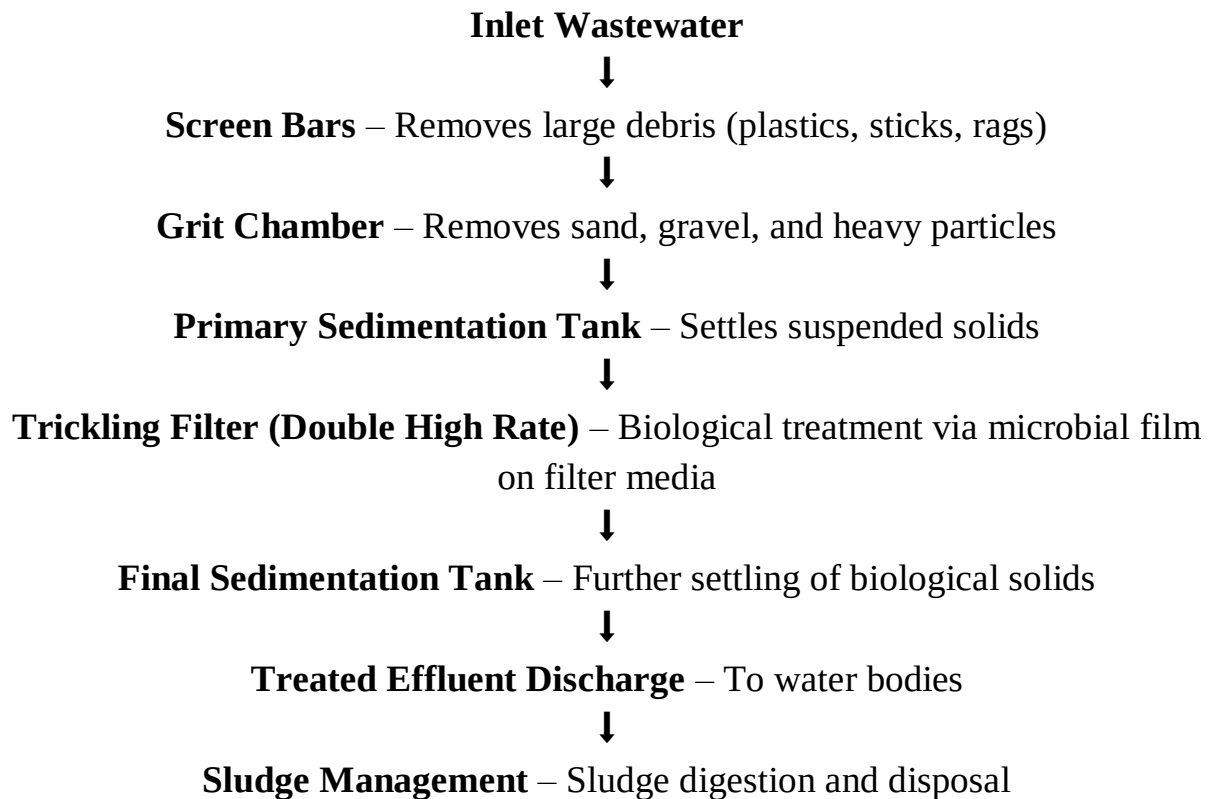
The process for the sanitary wastewater treatment in our project includes the following:

Preliminary treatment

Primary treatment

Secondary treatment

Flowchart: Wastewater Treatment Process



9.1 Preliminary Treatment

9.1.1 Screening

Screening is the first stage of wastewater treatment, where large solid materials such as plastics, leaves, and other debris are removed. This prevents clogging and damage to downstream treatment units.

Type: Racks or Bar Screens

Velocity through Screen = 0.8 m/s [coarse screen]

Vertical Projected Area = Flow rate/ velocity = 7.63 m²

Bar size = 10 mm * 50 mm

Spacing between bars = 50mm

Efficiency coefficient of bars = $50 / [50 + 10] = 5/6$

So, Vertical Projected Gross Area of screen = $5/6 * 7.63 = 6.35 \text{ m}^2$

Let, No. of Bars = 50

No. of openings = 51

Width of Screening channel = $50 * 10/1000 + 51 * 50/1000 = 3.05\text{m} = 3\text{m}$

Depth of flow = $6.35 / 3 = 2.11 \text{ m}$

Assuming top of screen to be 0.3m above the highest flow level and free board of 0.3 m

Total depth of screen channel = $2.71\text{m} = 3\text{m}$

Velocity of flow in screen channel = $6.108 / (3 * 3) = 0.678 \text{ m/s}$

Head loss = $0.0729(V^2 - v^2) = 0.013\text{m} = 13\text{mm}$, when screen cleaned.

When screen half clogged, Head loss = $0.0729(V^2 - v^2)$ but $V = 2 * 0.8$

$= 0.153\text{m} = 153\text{mm}$

Hence, the bar should be frequently cleaned to keep the head loss within a desired limit.

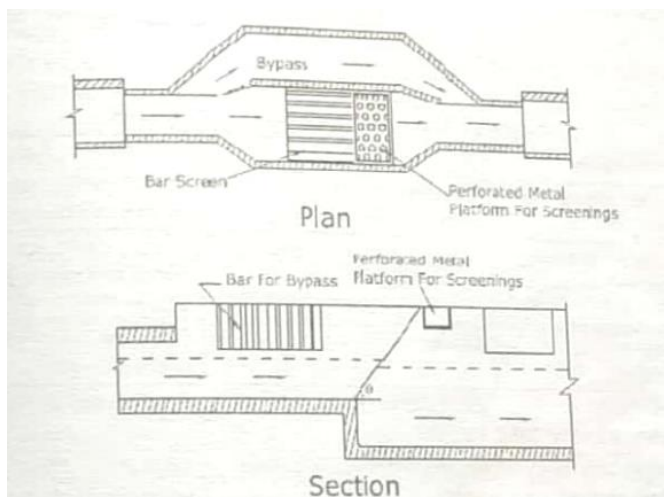


Figure 5-Bar Screen

Conclusion

- Number of bars = 50
- Bar width = 10mm
- Spacing between bars = 50 mm
- Bar Screen Area = $3\text{m} * 3\text{m}$
- Velocity in Screen Channel = 0.678 m/s

9.1.2 Grit Chamber

The grit chamber is designed to remove heavier inorganic particles like sand, gravel, and silt, which can cause wear and tear on equipment. Wastewater flows through the chamber at a controlled velocity, allowing grit to settle at the bottom while lighter organic matter continues to the next stage.

Total Sewage discharge (Q) = $6.10818 \text{ m}^3/\text{s}$

Let us assume,

Operating Temperature(T) = 27°C

Diameter of grit particles considered(d) = 0.25 mm

Specific gravity of grit (Sg)= 2.65

Specific gravity of organic matter (So) = 1.2

K=0.06

f=0.03

Settling velocity (Vs)= $60.6(Sg-1) * d*(3T+70)/100 = 0.038 \text{ m/s}$

Taking detention time (t)= 60 secs

Volume of grit chamber(V) = $Q*t = 6.10818*60 = 366.49 \text{ m}^3$

Depth of chamber (d) = $V_s * t = 0.038*60 = 2.28\text{m}$

Plan Area(A)= $V/d = 366.49/2.28 = 160.74 \text{ m}^2$

Provide 5 grit chambers.

Plan area of each grit chamber(A') = $160.74/5 = 32.148 \text{ m}^2$

Assume width of grit chamber(b) = 2 m

Length of grit chamber(l) = $A'/b = 16.07 \text{ m}$

Taking 0.25m for grit storage and 0.25m for free board, Total depth= $2.28+0.5=2.78\text{m}$

Check for critical velocity:

$$V_h = Q/(nbd) = 6.10818 / (5 * 2 * 2.78) = 0.22 \text{ m/s}$$

Self-cleansing velocity for grit:

$$V_{sc \text{ grit}} = \sqrt{(8k (S_g - 1) * d_g * g)/f}$$

$$= \sqrt{(8 * 0.06 * (2.65 - 1) * 0.25 * 10^{-3} * 9.81)/0.03} = 0.25 \text{ m/s.}$$

Self-cleansing velocity for organic matter:

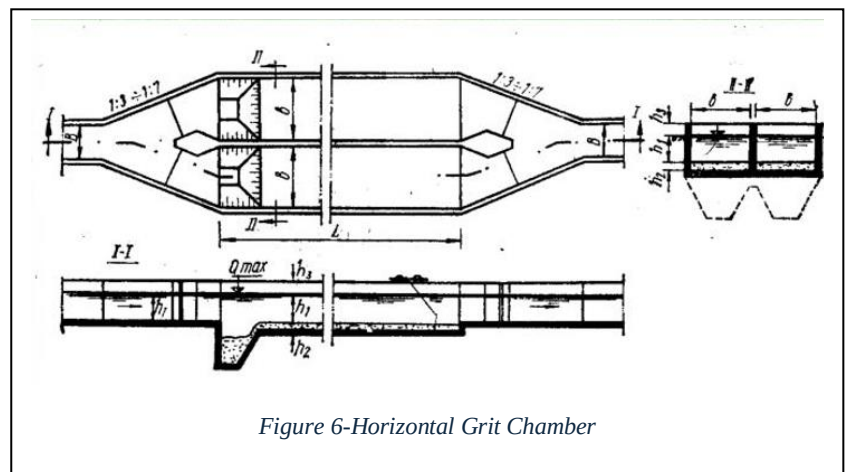
$$V_{sc \text{ organic matter}} = \sqrt{(8k (S_g - 1) * d_g * g)/f}$$

$$= \sqrt{(8 * 0.06 * (1.2 - 1) * 0.25 * 10^{-3} * 9.81)/0.03}$$

$$= 0.089 \text{ m/s.}$$

$$\text{As } V_h < V_{sc \text{ grit}} < V_{sc \text{ organic matter}}$$

So, OK.



Hence, Provide 5 parallel units of size 16.07 m*2m*2.78m. One unit is standby.

9.2 Primary Treatment

9.2.1 Sedimentation Tank

Sedimentation is defined as a process of removing fine suspended impurities that are heavier than water present in sewage under the action of gravitational force, this process is also known as settling or clarification. The sedimentation tanks are thus designed to remove fine suspended matter from the sewage effluent which are not removed by the grit chambers.

$$Q = 6.10818 \text{ m}^3/\text{s} = 527745.75 \text{ m}^3/\text{day}$$

Let detention time for sedimentation tank is 2 hours (1-4 hours)

Then,

Capacity or volume of tank(V) = $6.10818 \times 2 \times 3600 = 43978.89 \text{ m}^3$

Let assume average S.O. R = $40 \text{ m}^3/\text{m}^2/\text{day}$

Then surface area (A) = $Q / \text{S.O.R} = 21989.45 / 40 = 13193.66 \text{ m}^2$

Depth (D) = $V / A = 43978.89 / 13193.66 = 3.33 \text{ m}$

Let us provide 6 sedimentation units. Area of each unit = 2198.94 m^2

Assume $L/B=3$, then $L=3B$ $A=L \times B$ or $2198.94 = 3B^2$ or $B = 27.07 \text{ m}$

Hence $L=3 \times 27.07 = 81.22 \text{ m}$

Total depth of sedimentation tank = $D + \text{free board} + \text{sludge depth} = 3.33 + 0.3 + 1.0 = 4.63 \text{ m}$

Hence, provide 6 rectangular tanks having dimension length 81.22m, width 27.07 m and depth 4.63m.

9.3 Secondary Treatment

9.3.1 Trickling Filter

The trickling filter is a biological treatment unit where wastewater is distributed over a bed of media, such as rocks or plastic, to promote microbial growth. Microorganisms break down organic pollutants as wastewater trickles through the media, improving water quality. This process enhances biological treatment and reduces contaminants before the treated water is discharged or further processed.

Type = Two Stage Circular High-Rate Trickling Filter

BOD of raw sewage = 250 mg/ltr

BOD removal in the PST = 35 %

Organic loading of filter in 1st and 2nd stage = 10,000 kg/ha-m/day

Recirculation ratio for both stages = 1:4

Design discharge (Q) = 6.10818 cumecs = 527.75* 10⁶ ltr/day

For 1st stage TF:

Influent BOD = (1-0.35) *250

=162.5 mg/ltr

Re-circulation factor = $F_1 = [1+r]/[1+0.1r]^2$

= 1.1898

We have, Organic Loading = $U_1 = W_1/V_1F_1$

$V_1 = W_1/U_1F_1$

= 162.5*527.75 / 10,000*1.1898

= 7.21 ha-m

Assume Effective depth (d) = 2.5 m

Area (A_s) = 7.21/2.5 = 2.844 ha = 28440 m³

Let us provide 6 filters. Then, Plan area of each filter = 28440 / 6 = 4740 m³

Diameter (D_1) = $[4*A/ \pi]^{1/2} = 57.68m$

For Second Stage TF:

Let, C_o = effluent BOD from 1st stage = influent BOD for 2nd stage

C_e = effluent BOD for 2nd stage

C_i = influent BOD for 1st stage

Check for final effluent BOD:

From NRC equation,

$$[C_i - C_e]/C_i = 1/[1 + 0.0044\sqrt{U_1}]$$

$$[162.5 - C_e] / 162.5 = 1 / [1 + 0.0044\sqrt{10000}]$$

$$[162.5 - C_e] / 162.5 = 0.6944$$

Hence, $C_e = 49.652$ mg/litre

Also,

$$\frac{C_e - C'_e}{C_e} = \frac{1}{1 + \frac{0.0044\sqrt{U_2}}{1 - e}}$$

$$\frac{49.652 - C'_e}{49.652} = \frac{1}{1 + \frac{0.0044\sqrt{10000}}{1 - 0.6944}}$$

Hence, $C'_e = 29.3$ mg/litre < 35 mg/litre OK

Sizing of unit:

Again, we have $U_2 = W_2/V_2F_2$

$$V_2 = W_2/U_2F_2$$

$$= [49.653 * 527.74] / [10000 * 1.1898]$$

$$= 2.2 \text{ ha-m} = 22000 \text{ m}^3$$

Assuming Effective depth (d) = 2.5 m

$$\text{Area} = 22000/2.5 = 8800 \text{ m}^2$$

Providing 4 Second stage TF, Area of each = $A_s = 8800/4 = 2200 \text{ m}^2$

and diameter (D_2) = $[4 * A / \pi]^{1/2} = 52.92 \text{ m}$

Design Summary:1st stage TF:

Diameter= 57.68 m

Height= 2.5 m

Number of filters= 6

2nd stage TF:

Diameter= 52.92 m

Height= 2.5m

Number of filters= 4

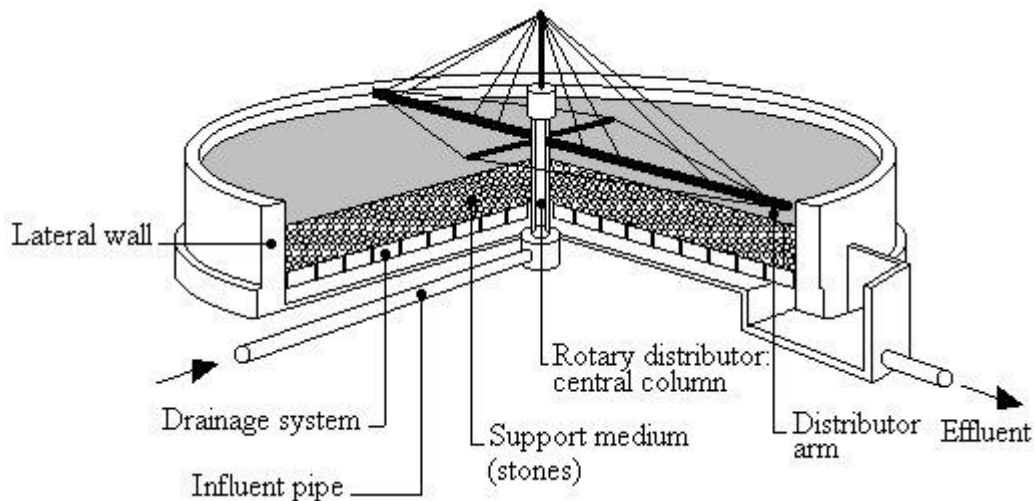


Figure 7-Circular Trickling Filter

(src:researchgate.net© [Marcos R. Vianna](#))

9.3.2 Secondary Sedimentation Tank

The effluent from the trickling filter undergoes further clarification in the secondary sedimentation tank to remove remaining suspended solids and biomass sloughed off during filtration. For design efficiency and uniformity, the secondary sedimentation tank is constructed with the same dimensions and structural specifications as the primary sedimentation tank. This standardization ensures consistency in treatment performance, facilitates maintenance, and optimizes sludge management within the wastewater treatment system.

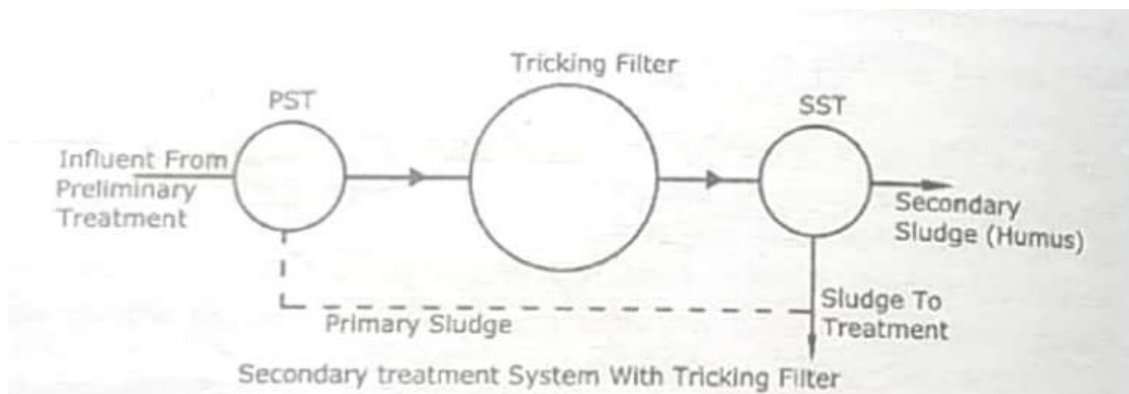


Figure 8-Secondary Sedimentation Tank System

10. SLUDGE TREATMENT AND DISPOSAL

Treated effluent and sludge are two products from sewage treatment plants. The treated effluent is directly disposed by dilution in water body or land treatment. However, the sludge that contains 0.25 to 12 percent solids, depending on treatment plant used for wastewater treatment needs further treatment before its disposal.

10.1 Disposal of effluent:

Disposal by Dilution

Here, the treated effluent is discharged into the treatment plan nearby primary watercourse, the Tal Khola. This river serves as the outlet of Rupa Lake, carrying its water downstream. The Tal Khola continues its course, eventually merging with the Seti River. By directing treated effluent into the Tal Khola, the sewage would undergo natural dilution and dispersion as it flows downstream, contributing to the self-purification processes before joining the Seti River.

It's essential to ensure that the discharged effluent meets environmental standards to protect the aquatic ecosystems and maintain water quality in both the Tal Khola and the Seti River. For this the raw sewage may be given partial treatment before being discharged into water body.

10.2 Sludge Disposal

In the sewage treatment system of Rupa Rural Municipality, sludge is generated at two key points: the primary sedimentation tank and the secondary sedimentation tank. The **primary sludge**, which consists of settleable solids from raw sewage, accumulates at the bottom of the primary sedimentation tank. Meanwhile, the **secondary sludge**, primarily composed of microbial biomass from the trickling filter, settles in the secondary sedimentation tank.

10.2.1 Sludge Weight Calculation

Primary Sedimentation Tank:

Flow rate= 527.75×10^6 ltr/day

Assume that primary treatment removes 30% BOD and 50% of suspended solids.

$BOD_5 = 250$ mg/l and $SS = 250$ mg/l.

Suspended solids removed in primary treatment = $0.5 \times 250 = 125$ mg/l

BOD_5 Influent entering in trickling filter plant = $(1-0.3) \times 250 = 175$ mg/l

Let solids production rate from trickling filter is at 0.5kg of BOD applied. Then, production of solids in trickling filter = $0.5 \times 175 = 87.5$ mg/l

Therefore, total solid production= $175+87.5 = 262.5$ mg/l = 138534 kg / day

10.2.2 Sludge Treatment

For the sludge production of 138,534 kg per day in Rupa Rural Municipality, an effective sludge treatment system is essential. Given the scale of sludge production, here are two treatment methods that could be considered:

1. Anaerobic Digestion

Anaerobic digestion is a biological treatment process that breaks down organic matter in the absence of oxygen. It is highly effective for treating large volumes of sludge. It produces biogas (methane), which can be used for energy generation, thus making it a sustainable and energy-efficient option. It also reduces the volume of sludge and minimizes odors.

- **Design Considerations:**
 - **Digester Tank:** The design of anaerobic digesters depends on the flow rate of sludge. The size of the tank should be large enough to accommodate the required daily input (138,534 kg). For effective treatment, retention times between 15 to 30 days are typically recommended.
 - **Temperature:** It can be designed for mesophilic (around 35-40°C) or thermophilic conditions (around 50-60°C). Thermophilic conditions speed up digestion but require more energy for heating.
 - **Biogas Recovery:** A gas recovery system would be necessary to collect and utilize the methane gas produced.
 - **Post-Treatment:** After digestion, the remaining sludge can be dewatered using centrifuges or belt presses to reduce the volume further.

2. Sludge Dewatering and Drying Beds

Dewatering is essential for reducing the volume of sludge before final disposal or further treatment. It removes water from the sludge and can be combined with drying beds or mechanical dewatering methods like belt filter presses. This method is cost-effective and suitable for rural municipalities with limited resources for advanced treatments like incineration.

- **Design Considerations:**
 - **Dewatering Units:** Centrifuges or belt filter presses can be used for efficient water removal. For large-scale treatment, a centrifuge with a high throughput capacity should be designed based on the daily sludge production.
 - **Drying Beds:** These are open-air systems where the dewatered sludge is spread out on a porous bed and dried under natural evaporation. The design must consider local climate conditions and the required drying period, which could take several weeks.
 - **Storage & Disposal:** The dried sludge can be stored for disposal or used as fertilizer if safe (low in pathogens and metals). However, if it is not suitable for land application, it can be disposed of in a landfill or further processed for incineration.

Composting is another viable option, particularly if organic matter from the sludge can be turned into valuable compost for agricultural use. This is an environmentally friendly process and can also reduce the volume of sludge.

- **Design Considerations:**

- **Aerobic Piles or Bins:** Sludge can be mixed with bulking agents (e.g., sawdust, wood chips) to improve aeration and accelerate the composting process.
- **Temperature Control:** The composting process requires temperature control to kill pathogens. The pile should reach temperatures of 55-65°C for several days to ensure pathogen destruction.
- **Aeration:** Piles should be turned into regular or use forced aeration systems to maintain aerobic conditions.
- **Maturation:** After the active composting phase, the sludge must mature for several weeks to stabilize and reduce pathogens.

10.2.3 Final Sludge Disposal

For Rupa Rural Municipality, the best sludge disposal method would be a combination of land application (if the treated sludge is safe for agriculture) and landfilling (for any residual or hazardous sludge).

1. **Land Application:** Treated sludge can be used as a soil conditioner, enriching agricultural land with nutrients. The sludge must undergo anaerobic digestion to reduce pathogens and heavy metals, ensuring safety. Design considerations include assessing soil nutrient needs, determining appropriate application rates, and regular monitoring of soil quality and crops to avoid overloading and contamination.
2. **Landfilling:** If land application is unsuitable, sludge can be disposed of in a controlled landfill. The sludge must be dewatered and stabilized through anaerobic digestion to reduce volume and leachate risks. The landfill design should include impermeable liners to prevent groundwater contamination, with systems for managing leachate and gas. Proper compaction and covering of the sludge are essential to minimize environmental impact.

Suggested Areas for Landfilling:

Higher altitudes or gently slope areas in the southwest part of Ward 5 (closer to rural, less populated areas) could also be considered, given the proximity to major roads and easy access.

A site near the periphery of Ward 6, closer to the Seti River, but at a safe distance from it, could be a good option, provided the land is away from the river's flood zone.

A site survey to assess these factors more closely should be conducted and environmental and regulatory standards for landfilling should be ensured.

11. SUMMARY OF DESIGN

Design of Sewer

	Main Sewer Line A	Sub-Main Sewer Line B B1 B2 B3			Sub-Main Sewer Line C C1 C2	
Material Used	Concrete	Concrete			Concrete	
Slope	0.004	0.003			0.003	
Diameter	2000mm	1200mm	1400mm	1500mm	1200mm	1400mm
Length	1272.63 m	2575.21 m	2487.48 m	2060.278 m	1937.467 m	1334.692 m

Treatment Design:

Preliminary Treatment	
Screening	
Number of bars = 50	
Bar width = 10mm	
Spacing between bars = 50 mm	
Bar Screen Area = 3m * 3m	
Velocity in Screen Channel = 0.678 m/s	
Grit Chamber	
Units	5 + 1 standby
Dimension	6.07m * 2m* 2.78m

Primary Treatment	
Primary Sedimentation Tank	
Units	6
Dimension	81.22m*27.07 m * 4.63m.
Secondary Treatment	
Trickling Filter	
Type = Two Stage Circular High-Rate Trickling Filter	
1st stage TF:	2nd stage TF:
Diameter= 57.68 m	Diameter= 52.92 m
Height= 2.5 m	Height= 2.5m
Number of filters= 6	Number of filters= 4
Secondary Sedimentation Tank	
Units	6
Dimension	81.22m*27.07 m * 4.63m.

12. OPERATION AND MAINTENANCE

Due to the municipality's rural and hilly terrain, maintaining an efficient sewerage system is challenging but essential to prevent water pollution, safeguard public health, and sustain local water bodies. Regular operation and maintenance (O&M) practices are necessary to prevent system failures and ensure long-term sustainability.

General Problems in Sewerage System

The key challenges in managing the sewerage system in Rupa Rural Municipality include:

- **Blockages & Sedimentation:** Due to inadequate slope in pipelines, silt deposition, and household waste disposal.
- **Leakages & Structural Damage:** Aging pipes, improper joint connections, and landslides cause pipe breakages.
- **Sewer Overflows & Waterlogging:** During monsoons, heavy rainfall increases water inflow, leading to overflow issues.
- **Lack of Proper Treatment Facilities:** Limited wastewater treatment infrastructure results in direct discharge into nearby streams.
- **Community Awareness Issues:** Improper waste disposal, such as throwing plastics and solid waste into drains, worsens system efficiency.

Recommended Maintenance Works

Regular cleaning and inspection of sewer lines through desilting, flushing, and monitoring to prevent blockages. Immediate repair of leaks and structural damages, especially in landslide-prone areas, to maintain pipeline integrity. Stormwater management by constructing proper drainage channels and monitoring open drains during monsoons to prevent overflows. Improving wastewater treatment through decentralized facilities and eco-friendly disposal methods like constructed wetlands. Community awareness programs to promote proper waste disposal, septic tank maintenance, and solid waste segregation. These measures will enhance system efficiency, prevent contamination, and improve sanitation in the municipality.

13. COST ESTIMATION

Construction Materials

Locally sourced materials such as sand, aggregate, and timber have been priced based on collection and transportation costs. Non-local materials will be procured from the nearest market, with rates adopted from district-approved rates, including transportation costs.

Rate Analysis

Unit rates for construction items, including sewer lines and treatment units, are determined based on the Government of Nepal (GON) norms. Where district rates are unavailable, market rates from suppliers and contractors are utilized. Overhead costs include 15% for the contractor's profit, and VAT is added separately.

Overhead, Profit, and Taxes

An overhead and contractor's profit of 15% is included in all unit rates. VAT of 13% has been added separately in the final cost summary.

Contingency

A contingency of 10% of the total cost is included to account for price escalation and changes in physical quantities during construction.

Summary of Cost Estimates

The total cost estimation is based on quantity take-offs and unit rates as per the rate analysis, following the Procurement Act, 2063, and Procurement Regulation, 2064.

Cost Breakdown:

S.N.	Description of Work	Amount (NRs)
1	General Items	10,00,000.00
2	Sewer Line Construction	5,00,00,000.00
3	Manholes & Inspection Chambers	50,00,000.00
4	Wastewater Treatment Unit	2,00,00,000.00
5	Sludge Drying Beds	30,00,000.00
6	Outfall & Disposal System	5,00,000.00
	Subtotal	7,95,00,000.00
	Contingency (10%)	79,50,000.00
	Subtotal	87,450,000.00
	VAT (13%)	11,368,500.00
	Grand Total	9,88,18,500.00

Note: Actual costs may vary based on site conditions, material availability, and market fluctuations. This is the basic estimation with low accuracy.

This estimate provides an overview of the total costs associated with the sanitation project, ensuring efficiency and compliance with government norms.

14. CONCLUSION

The sanitation project for Rupa Rural Municipality, Wards 5 and 6, was designed to address critical issues in sewage management, ensuring proper wastewater disposal and improved hygiene. The study identified inadequate drainage, improper waste disposal, and a lack of sanitation facilities as key challenges. To mitigate these, the project proposes a structured sewage system with appropriate pipelines, treatment and disposal ideas for better waste management. The design aligns with the region's topography and population needs, incorporating sustainable and cost-effective solutions to enhance long-term efficiency.

For future extensions, the project can integrate advanced treatment facilities, decentralized wastewater management, and rainwater harvesting to optimize resource utilization. Regular maintenance, government support, and active community participation will be crucial in ensuring the system's effectiveness. The findings highlight the necessity of a well-planned and continuously monitored sanitation framework to create a healthier and more sustainable living environment for the residents of Rupa Rural Municipality.

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